Qingyang Zhou

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https://esperarr.github.io/CV/

EDUCATION

Tsinghua University September 2022 - Present
Xinya College, Tsinghua University: https://www.xyc.tsinghua.edu.cn/index.htm
Department of Computer Science and Technology, Tsinghua University: https://www.cs.tsinghua.edu.cn/
GPA 3.6/4.0
Selected A+ professional compulsory course: Introduction to Complex Analysis, Media Computing, Database Special Topic Training;
Selected A/A- professional compulsory courses: Linear Algebra, Fundamentals of Programming, Programming and Training, Software Engineering, Introduction to Artificial Intelligence, Introduction to Machine Learning, Digital Logic Experimentation, Computer Network Security Technology, Fundamentals of Computer Graphics, Network Security Engineering and Practice

RESEARCH PUBLICATIONS

"Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation"
Yuwei Zhou, Xin Wang, Hong Chen, Yipeng Zhang, ***Qingyang Zhou****, Wenwu Zhu*
Submitted for publication
OpenReview preprint

"Towards Mutually Illuminative Collaborative Human–AI Deliberate Discussion"
Zipeng Zhang , Shiwei Wu, Ran Ran, ***Qingyang Zhou****, Ranrui Ma, Jiaye Leng, Jian Zeng, Zhenhui, Chun Yu*
Submitted for publication

INTERNSHIP / RESEARCH

Client Development Engineer Intern at Tencent (Project MAX) May 2026 - August 2026
Tencent, Project MAX Team
- Optimized NavMesh pathfinding to improve agent movement smoothness, resolving instability on terrain bumps and convex corner edges.
- Implemented NPC auto-navigation and pathfinding behavior using custom FixNavMesh algorithms.
- Developed a GM panel for debugging and performance stats toggling in the battle UI.
- Built and refined game control panel UI for the 2V2 battle mode.
- Also contributed to various other client features and gameplay systems across the project.

Scientific Research Intern at Multimedia and Network Big Data Lab July 2023 - July 2024
Tsinghua University Multimedia and Network Big Data Lab: mn.cs.tsinghua.edu.cn
- Contributed to the research project "Video Generation System Combining Large Language Models with Diffusion Models."
- Studied diffusion models and personalized image generation.
- Assisted with the research and manuscript preparation for "Curriculum Multi-Reward Reinforcement Learning for Customized Text-to-Image Generation."

Scientific Research Intern at Pervasive Human-Computer Interaction Lab January 2025 - Present
Tsinghua University Pervasive Human-Computer Interaction Laboratory: pi.cs.tsinghua.edu.cn
- Conducted research on aligning human and AI thought processes.
- Developed a human-AI interactive writing assistant to support collaborative idea generation.

· Contributed to the research and manuscript preparation for "Towards Mutually Illuminative Collaborative Human–AI Deliberate Discussion" (submitted for publication).

Scientific Research Intern at Information Retrieval Lab October 2024 - Present
Information Retrieval Lab at Tsinghua University: thuir.cn
· Focused on improving traditional recommendation models (such as FM, DeepFM, AFM).
· Incorporated SAE plugins into these models to enhance their interpretability, controllability, and transparency.

POSITIONS OF RESPONSIBILITY

Secretary of the Intelligent Agent Department of the Student Science and Technology Association February 2023 - Present
Tsinghua University Computer Science and Technology Department, Student Science and Technology Association, Intelligent Agent Department: net9.org/home
· Participated in the game development and event organization for the Intelligent Agent Competition.
· Mainly responsible for the development of the Intelligent Agent-Based Game 'Generals' Project in the 28th Tsinghua University Intelligent Agent Competition.

Tutor of Tsinghua University Science and Engineering Summer Camp July 2024 - August 2024
Tsinghua University 2024 Science and Engineering Summer Camp, I6 Class Tutor
· Responsible for event organization, answering professional questions, and other related tasks.
· Led and organized the operation of certain engineering-related activity projects, such as intelligent robot design and card-bridge construction.

Volunteer Tutor at Tsinghua Q&A Workshop September 2025 - Present
Tsinghua University, Center for Student Learning and Development: learning.tsinghua.edu.cn
· Provided tutoring for programming and computer science courses, including C/C++ programming, data structures, and algorithms.
· Assisted students with debugging, algorithm design, and understanding core programming concepts during regular tutoring sessions.

SKILLS

Languages	Chinese (native), English (fluent), Spanish (basic)
Programming	C/C++, Python, Java, JavaScript/TypeScript, Rust, Verilog/SystemVerilog
Graphics & Game Development	OpenGL, GLSL, Unity, path tracing, rendering pipelines
AI / Machine Learning	PyTorch, NumPy, machine learning and deep learning fundamentals
Tools	Git, Linux, Docker, LaTeX, Markdown

PROJECTS

Intelligent Agent-Based Game "Generals" September 2023 - May 2024
Major Project in the 28th Tsinghua University Intelligent Agent Competition
· A turn-based strategy game available at saiblo.net/game/35.
· Participated in the development of this game project.
· Responsible for backend logic, unit tests, debugging, SDK development, documentation, and guiding participants through the competition.

Path Tracing Renderer June 2026
Computer Graphics Course Project
· Source code: github.com/Esperarr/TinyPathTracer
· A physically-based path tracer supporting both Whitted-style ray tracing and Monte Carlo path tracing with Russian Roulette for unbiased termination.

· Implemented PBR materials with GGX microfacet importance sampling (isotropic and anisotropic), Cook-Torrance BRDF, and Fresnel Schlick, covering diffuse, mirror, glass, and glossy surfaces.
· Added BVH acceleration for 100k+ face meshes, NEE + MIS with power heuristic weighting, depth of field, MSAA anti-aliasing, gamma correction, and OpenMP parallelization.

Text Sentiment Analysis Model May 2024
Course Project
· A text sentiment classification model that can analyze the sentiment of the corpus based on the input text.
· Built CNN, RNN, and MLP sentiment classifiers in Python and PyTorch, then evaluated their performance. The final model achieved over 90% accuracy.

Five-Stage Pipeline CPU November 2025 - December 2025
Computer Organization Course Project
· A five-stage pipeline CPU hardware program capable of executing 19 basic RISC-V instructions as well as some extended instructions.
· Wrote hardware code in System Verilog, implementing functions such as data passing, signal control, and hazard handling in a five-stage pipeline, and completed the debugging process using Vivado simulation.

Connect Four AI June 2024
Course Project
· A Connect Four AI agent developed in Python.
· Implemented Monte Carlo Tree Search (MCTS) with Upper Confidence bounds for Trees (UCT), achieving a win rate above 90% against baseline AI opponents.

Chinese Pinyin Input Method April 2024
Course Project
· A Python input-method program that converts Pinyin into Chinese characters.
· Used a hidden Markov model and the Viterbi algorithm, achieving over 85% character prediction accuracy.

Android News App September 2023
Java Course Major Project
· An Android news app written in Java that fetches backend data and displays it in a categorized, swipeable interface.
· Implemented pull-to-refresh, pagination, video playback, tabs, and article like/favorite features; independently completed the app architecture, implementation, and debugging in Android Studio.

EXTRA-CURRICULAR

- Served as a volunteer for Tsinghua University's Tianjin recruitment team August 2023
- Served as the Publicity Department Executive in Xinya College. September 2022 - September 2023